

PRACTICAL WEEK-9

Task-1

Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>
#include <time.h>

#define BUFFER_SIZE 4096

void low_level_copy(const char *source, const char *destination)
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    src = open(source, O_RDONLY);

    if (src < 0)
    {
        perror("Error opening source file");
        exit(1);
    }

    dest = open(destination, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest < 0)
    {
        perror("Error opening destination file");
        close(src);
        exit(1);
    }

    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read)
```

```

start = clock();

low_level_copy(source, "low_level_copy.txt");

end = clock();

low_time = ((double)(end - start)) / CLOCKS_PER_SEC;

printf("Low-level copy completed.\n");
printf("Low-level time: %f seconds\n\n", low_time);

printf("Starting standard library file copy...\n");

start = clock();

library_copy(source, "library_copy.txt");

end = clock();

library_time = ((double)(end - start)) / CLOCKS_PER_SEC;

printf("Standard library copy completed.\n");
printf("Standard library time: %f seconds\n\n", library_time);

printf("Performance comparison:\n");

if (low_time < library_time)
    printf("Low-level I/O was faster in this run.\n");
else if (library_time < low_time)
    printf("Standard library I/O was faster in this run.\n");
else
    printf("Both took approximately the same time.\n");

return 0;
}

```

```

laasya@laasya:~$ mkdir filecopy
laasya@laasya:~$ cd filecopy
laasya@laasya:~/filecopy$ nano filecopy.c
laasya@laasya:~/filecopy$ nano input.txt
laasya@laasya:~/filecopy$ gcc filecopy.c -o filecopy
laasya@laasya:~/filecopy$ ./filecopy
Starting low-level file copy...
Low-level copy completed.
Low-level time: 0.000125 seconds

Starting standard library file copy...
Standard library copy completed.
Standard library time: 0.000124 seconds

Performance comparison:
Standard library I/O was faster in this run.
laasya@laasya:~/filecopy$ ls
filecopy  filecopy.c  input.txt  library_copy.txt  low_level_copy.txt
laasya@laasya:~/filecopy$ cat low_level_copy.txt
This is a test file for low-level Linux file I/O.
This file will be copied using open, read, write and close.
laasya@laasya:~/filecopy$ |

```

TASK 2: Develop a program that redirects standard input and output using file descriptors and dup(). Demonstrate how shells implement I/O redirection.

```

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input_fd, output_fd;

    input_fd = open("input.txt", O_RDONLY);

    if (input_fd < 0)
    {
        perror("Error opening input file");
        return 1;
    }

    output_fd = open("output.txt",
                    O_WRONLY | O_CREAT | O_TRUNC,
                    0644);

    if (output_fd < 0)
    {
        perror("Error opening output file");
        close(input_fd);
        return 1;
    }

    /*
     * Redirect standard input (stdin)
     * to input.txt
     */
    if (dup2(input_fd, STDIN_FILENO) < 0)
    {
        perror("dup2 input");
        return 1;
    }
}

```

```

    if (output_fd < 0)
    {
        perror("Error opening output file");
        close(input_fd);
        return 1;
    }

    /*
     * Redirect standard input (stdin)
     * to input.txt
     */
    if (dup2(input_fd, STDIN_FILENO) < 0)
    {
        perror("dup2 input");
        return 1;
    }

    /*
     * Redirect standard output (stdout)
     * to output.txt
     */
    if (dup2(output_fd, STDOUT_FILENO) < 0)
    {
        perror("dup2 output");
        return 1;
    }

    close(input_fd);
    close(output_fd);

    printf("This message is redirected to output.txt\n");

    printf("Standard input and output have been redirected.\n");

    return 0;
}

```

```

laasya@laasya:~/filecopy$ ls
filecopy  filecopy.c  input.txt  library_copy.txt  low_level_copy.txt
laasya@laasya:~/filecopy$ cat low_level_copy.txt
This is a test file for low-level Linux file I/O.
This file will be copied using open, read, write and close.
laasya@laasya:~/filecopy$ nano redirect.c
laasya@laasya:~/filecopy$ gcc redirect.c -o redirect
laasya@laasya:~/filecopy$ ./redirect
laasya@laasya:~/filecopy$ cat output.txt
This message is redirected to output.txt
Standard input and output have been redirected.
laasya@laasya:~/filecopy$ cat input.txt > shell_output.txt
laasya@laasya:~/filecopy$ cat shell_output.txt
This is a test file for low-level Linux file I/O.
This file will be copied using open, read, write and close.
laasya@laasya:~/filecopy$

```